SSC Maharashtra Board — Algebra (Mathematics Part I)

March 2026 — Model Solutions

Time: 2 Hours | Max. Marks: 40

Q.1

Q.1 (A) Choose the correct answer and write the alphabet of it in front of the subquestion number: [4 Marks]

(i) Rate of GST on essential commodities is ...

Essential commodities such as grains, milk, salt, etc. are exempted from GST.
$\therefore$ The rate of GST on essential commodities is 0%.

$$\boxed{\text{Ans: (C) } 0\%}$$

(ii) Out of the following equations, which one is *not* a quadratic equation?

Check option (A): $x^2 + 4x = 11 + x^2$
$\therefore x^2 + 4x - x^2 = 11$
$\therefore 4x = 11$
This is a linear equation, not a quadratic equation.

$$\boxed{\text{Ans: (A) } x^2 + 4x = 11 + x^2}$$

(iii) To draw graph of $4x + 5y = 19$, find the value of y when $x = 1$.

Substituting $x = 1$ in $4x + 5y = 19$:

$$4(1) + 5y = 19 \implies 5y = 15 \implies y = 3$$

$$\boxed{\text{Ans: (B) } y = 3}$$

(iv) Sum of first five multiples of 3 is ...

First five multiples of 3 are: 3, 6, 9, 12, 15.
Sum $= 3 + 6 + 9 + 12 + 15 = 45$

$$\boxed{\text{Ans: (A) } 45}$$

Q.1 (B) Solve the following subquestions: [4 Marks]

(i) On a certain article if the rate of CGST is 9%, then what is the rate of GST?

GST = CGST + SGST

For intra-state transactions, CGST = SGST.

$\therefore$ SGST = 9%

$\therefore$ Rate of GST = 9% + 9%

$$\boxed{\text{Rate of GST} = 18\%}$$

(ii) For simultaneous equations in variables x and y, if $D_x = -42$ and $D = -3$, then find the value of x.

By Cramer's rule:

$$x = \frac{D_x}{D} = \frac{-42}{-3} = 14$$

$$\boxed{x = 14}$$

(iii) When two coins are tossed simultaneously, write the sample space 'S'.

When two coins are tossed simultaneously, the possible outcomes are:

$$\boxed{S = \{HH, HT, TH, TT\}}$$

(iv) Write the following quadratic equation in the form $ax^2 + bx + c = 0$:
$2y = 10 - y^2$

$2y = 10 - y^2$

$\therefore y^2 + 2y = 10$

$\therefore y^2 + 2y - 10 = 0$

Comparing with $ay^2 + by + c = 0$:

$$\boxed{a = 1, \quad b = 2, \quad c = -10}$$

Q.2

Q.2 (A) Complete and write any *two* activities from the following: [4 Marks]

(i) If one die is rolled, then find the probability of the event, that the number on the upper face is prime.

Solution:

'S' is the sample space.

$S = \{1, 2, 3, 4, 5, 6\}$
$\therefore n(S) = \boxed{6}$

Event A: Prime number on the upper face.
$A = \{2, 3, 5\}$
$\therefore n(A) = \boxed{3}$

$\therefore P(A) = \frac{n(A)}{\boxed{\text{n(S)}}} = \frac{\boxed{3}}{6}$

$$\boxed{P(A) = \frac{1}{2}}$$

(ii) Write the correct number in the given boxes from the given A.P.: 70, 60, 50, 40, . . .

Here $t_1 = \boxed{70}$, $t_2 = 60$, $t_3 = \boxed{50}$
$\therefore a = \boxed{70}$, $d = t_2 - t_1 = 60 - 70 = \boxed{-10}$

(iii) Complete the following table to draw graph of the equation $x + y = 3$.

From $x + y = 3 \implies y = 3 - x$

x	$\boxed{0}$	3	$\boxed{-2}$
y	$\boxed{3}$	$\boxed{0}$	5
(x, y)	$\boxed{(0, 3)}$	$\boxed{(3, 0)}$	$\boxed{(-2, 5)}$

Explanation:

- When $x = 3$: $y = 3 - 3 = 0$
- When $y = 5$: $x = 3 - 5 = -2$
- When $x = 0$: $y = 3 - 0 = 3$

Q.2 (B) Solve any *four* subquestions from the following: [8 Marks]

(i) If two coins are tossed simultaneously, find the probability of the event, "getting at least one head."

When two coins are tossed simultaneously:
$S = \{HH, HT, TH, TT\}$
$\therefore n(S) = 4$

Event A: Getting at least one head.
$A = \{HH, HT, TH\}$
$\therefore n(A) = 3$

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{4}$$

$$\boxed{P(A) = \frac{3}{4}}$$

(ii) Solve the following simultaneous equations:
$x + y = 4$; $2x - y = 2$

$$x + y = 4 \quad \ldots(\mathrm{I}) \tag{1}$$
$$2x - y = 2 \quad \ldots(\mathrm{II}) \tag{2}$$

Adding equations (I) and (II):

$$x + y + 2x - y = 4 + 2$$

$$3x = 6 \implies x = 2$$

Substituting $x = 2$ in equation (I):

$$2 + y = 4 \implies y = 2$$

$$\boxed{x = 2, \quad y = 2}$$

(iii) Solve the following quadratic equation by factorisation method:
$x^2 + x - 20 = 0$

$x^2 + x - 20 = 0$
We need two numbers whose product is -20 and sum is 1.
Those numbers are 5 and -4.
$\therefore x^2 + 5x - 4x - 20 = 0$
$\therefore x(x + 5) - 4(x + 5) = 0$
$\therefore (x + 5)(x - 4) = 0$
$\therefore x + 5 = 0 \quad \text{or} \quad x - 4 = 0$
$\therefore x = -5 \quad \text{or} \quad x = 4$

$$\boxed{x = -5 \quad \text{or} \quad x = 4}$$

(iv) Given A.P. is 12, 16, 20, 24, . . . Find the 24th term of this progression.

Here $a = 12$, $d = 16 - 12 = 4$, $n = 24$
We know $t_n = a + (n - 1)d$
$\therefore t_{24} = 12 + (24 - 1) \times 4$
$\therefore t_{24} = 12 + 23 \times 4$
$\therefore t_{24} = 12 + 92$

$$\boxed{t_{24} = 104}$$

(v) If $\sum x_i f_i = 1265$ and $N = 50$, then find the mean of the given data.

We know:

$$\bar{x} = \frac{\sum x_i f_i}{N} = \frac{1265}{50}$$

$$\boxed{\bar{x} = 25.3}$$

Q.3

Q.3 (A) Complete and write any *one* activity from the following: [3 Marks]

(i) Shreekar bought a Laptop with 10% discount on printed price. The printed price of that Laptop was Rs. 50,000. 18% GST was charged on discounted price. Find the amount of CGST and SGST. What amount did Shreekar pay?

Solution:

Discount = 10% of 50,000 = Rs. $\boxed{5{,}000}$

$\therefore$ Taxable value of Laptop = 50,000 − $\boxed{5{,}000}$

= Rs. 45,000

$\therefore$ Rate of GST = $\boxed{18\%}$

$\therefore$ Rate of CGST = 9%

$\therefore$ CGST = 9% of 45,000 = $\boxed{\dfrac{9}{100}} \times 45{,}000$

= Rs. $\boxed{4{,}050}$

$\therefore$ SGST = Rs. 4,050

Total amount of Laptop = 45,000 + 4,050 + 4,050 = Rs. $\boxed{53{,}100}$

$\boxed{\text{Shreekar paid Rs. } 53{,}100}$

(ii) One of the roots of the quadratic equation $5m^2+2m+k=0$ is $\frac{-7}{5}$. Complete the following activity to find the value of 'k'.

$\frac{-7}{5}$ is a root of quadratic equation $5m^2 + 2m + k = 0$.

$\therefore$ put $m = \frac{-7}{5}$ in the equation.

$$\therefore 5 \times \boxed{\left(\frac{-7}{5}\right)}^2 + 2 \times \boxed{\left(\frac{-7}{5}\right)} + k = 0$$

$$\therefore 5 \times \frac{49}{25} + 2 \times \frac{-7}{5} + k = 0$$

$$\therefore \boxed{\frac{49}{5}} + \boxed{\frac{-14}{5}} + k = 0$$

$$\therefore \frac{49-14}{5} + k = 0$$

$$\therefore \boxed{\frac{35}{5}} + k = 0$$

$\therefore 7 + k = 0$

$\therefore k = \boxed{-7}$.

$\boxed{k = -7}$

Q.3 (B) Solve any *two* subquestions of the following: [6 Marks]

(i) Two numbers differ by 3. The sum of twice the smaller number and thrice the greater number is 19. Find the numbers.

Let the smaller number be x and the greater number be y.

According to the first condition:

$$y - x = 3 \quad \ldots \text{(I)} \tag{3}$$

According to the second condition:

$$2x + 3y = 19 \quad \ldots \text{(II)} \tag{4}$$

From equation (I): $y = x + 3$
Substituting in equation (II):
$2x + 3(x + 3) = 19$
$\therefore 2x + 3x + 9 = 19$
$\therefore 5x = 10$
$\therefore x = 2$

$\therefore y = x + 3 = 2 + 3 = 5$

$$\boxed{\text{The smaller number is 2 and the greater number is 5.}}$$

(ii) Sachin invested some amount in a National Saving Certificate Scheme. In the first year he invested Rs. 5,000, in the second year Rs. 7,000, in the third year Rs. 9,000 and so on. Find the total amount that he invested in 12 years.

The investments form an A.P.: 5000, 7000, 9000, ...
Here $a = 5000$, $d = 7000 - 5000 = 2000$, $n = 12$
Total amount invested $= S_n = \dfrac{n}{2}[2a + (n-1)d]$

$$S_{12} = \frac{12}{2}[2 \times 5000 + (12 - 1) \times 2000]$$

$$= 6[10000 + 11 \times 2000]$$

$$= 6[10000 + 22000]$$

$$= 6 \times 32000$$

$$\boxed{S_{12} = \text{Rs. } 1{,}92{,}000}$$

(iii) There are six cards in a box, each bearing a number from 0 to 5. Find the probability of each of the following events, that:
(1) The number on the card drawn is a natural number.
(2) The number on the card drawn is a whole number.

The cards bear numbers: 0, 1, 2, 3, 4, 5.
$S = \{0, 1, 2, 3, 4, 5\}$
$\therefore n(S) = 6$

(1) Event A: The number on the card drawn is a natural number.
Natural numbers from 0 to 5 are: 1, 2, 3, 4, 5.
$A = \{1, 2, 3, 4, 5\}$
$\therefore n(A) = 5$

$$P(A) = \frac{n(A)}{n(S)} = \frac{5}{6}$$

$$\boxed{P(A) = \frac{5}{6}}$$

(2) Event B: The number on the card drawn is a whole number.
Whole numbers from 0 to 5 are: 0, 1, 2, 3, 4, 5.
$B = \{0, 1, 2, 3, 4, 5\}$
$\therefore n(B) = 6$

$$P(B) = \frac{n(B)}{n(S)} = \frac{6}{6}$$

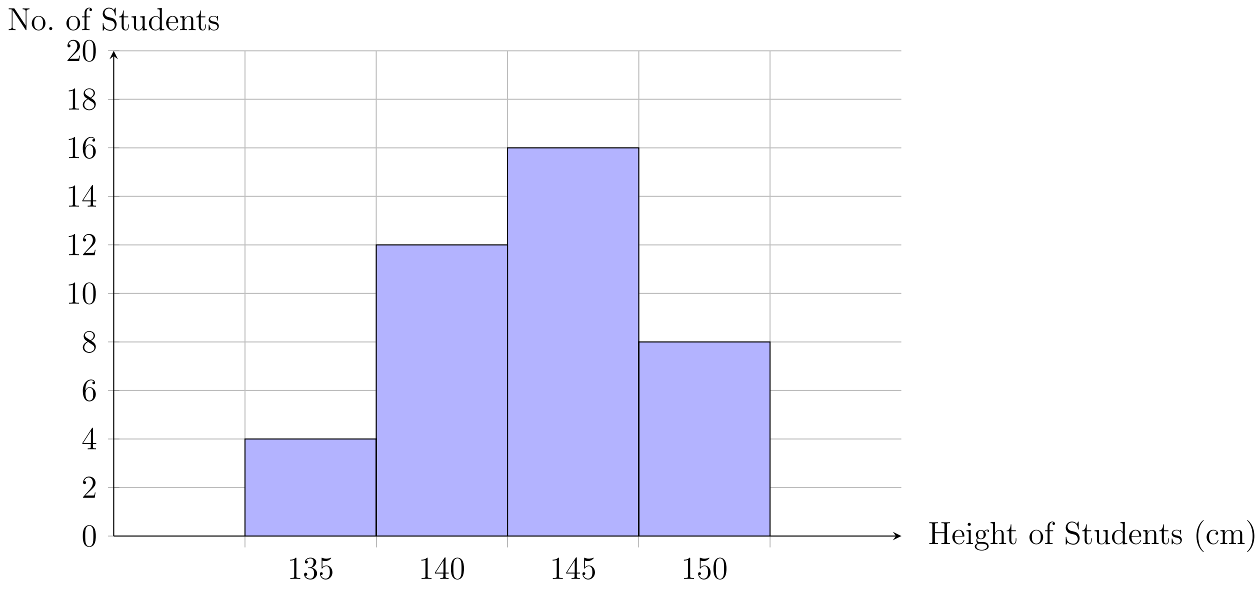
$$\boxed{P(B) = 1}$$

(iv) Draw a histogram of the following data:

Height of Students (cm)	**No. of Students**
135–140	4
140–145	12
145–150	16
150–155	8

Scale:

- X-axis: 1 cm = 5 cm (Height)
- Y-axis: 1 cm = 2 students



Q.4 Solve any *two* subquestions of the following: [8 Marks]

(i) The sum of two roots of a quadratic equation is 7 and sum of their cubes is 91, find the equation.

Let the roots be α and β.

Given: $\alpha + \beta = 7$ and $\alpha^3 + \beta^3 = 91$

We know: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
$\therefore 91 = (7)^3 - 3\alpha\beta(7)$
$\therefore 91 = 343 - 21\alpha\beta$
$\therefore 21\alpha\beta = 343 - 91 = 252$
$\therefore \alpha\beta = \dfrac{252}{21} = 12$
The quadratic equation with roots α and β is:
$x^2 - (\alpha + \beta)x + \alpha\beta = 0$
$\therefore x^2 - 7x + 12 = 0$

$$\boxed{x^2 - 7x + 12 = 0}$$

Verification: $x^2 - 7x + 12 = (x - 3)(x - 4) = 0$, so roots are 3 and 4.
Sum $= 3 + 4 = 7$ ✓
Sum of cubes $= 27 + 64 = 91$ ✓

(ii) Smt. Tejaswini purchased 100 shares of FV Rs. 80 when the MV is Rs. 900. She paid brokerage at the rate of 0.5% and 18% GST on brokerage. Find the following:
(1) Net amount paid for 100 shares.
(2) Brokerage paid on sum invested.
(3) GST paid on brokerage.
(4) Total amount paid for 100 shares.

Given: FV = Rs. 80, MV = Rs. 900, Number of shares = 100
Rate of brokerage = 0.5%, GST on brokerage = 18%

(1) Net amount paid for 100 shares (sum invested):

$$\text{Sum invested} = \text{Number of shares} \times \text{MV} = 100 \times 900$$

$$\boxed{\text{Sum invested} = \text{Rs.}\,90{,}000}$$

(2) Brokerage paid on sum invested:

$$\text{Brokerage} = 0.5\% \text{ of } 90{,}000 = \frac{0.5}{100} \times 90{,}000$$

$$\boxed{\text{Brokerage} = \text{Rs.}\,450}$$

(3) GST paid on brokerage:

$$\text{GST} = 18\% \text{ of } 450 = \frac{18}{100} \times 450$$

$$\boxed{\text{GST on brokerage} = \text{Rs.}\,81}$$

(4) Total amount paid for 100 shares:

$$\text{Total amount} = \text{Sum invested} + \text{Brokerage} + \text{GST}$$

$$= 90{,}000 + 450 + 81$$

$$\boxed{\text{Total amount paid} = \text{Rs.}\,90{,}531}$$

(iii) The following table shows the funds collected by 105 students for flood affected people. Find the median of the data.

Fund (Rupees)	**No. of Students (f)**	**Cumulative Freq. (cf)**
10–20	10	10
20–30	37	47
30–40	15	62
40–50	20	82
50–60	7	89
60–70	8	97
70–80	8	105
Total	$N = 105$	

$\dfrac{N}{2} = \dfrac{105}{2} = 52.5$
The cumulative frequency just greater than 52.5 is 62.
$\therefore$ The median class is 30–40.
Here: $L = 30,\ f = 15,\ cf = 47,\ h = 10$

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{52.5 - 47}{15} \times 10$$

$$= 30 + \frac{5.5}{15} \times 10 = 30 + \frac{55}{15} = 30 + 3.67$$

$$\boxed{\text{Median} \approx 33.67}$$

Q.5 Solve any *one* subquestion of the following: [3 Marks]

(i) A survey was conducted in the school. Students using different modes of transport is given below:

Mode of Transport	**Percentage of Students**
Bicycle	35
Bus	a
Railway	$2a$
Rickshaw	20

Find the value of a and draw the Pie diagram.

Step 1: Find the value of a.
Total percentage = 100%
$\therefore 35 + a + 2a + 20 = 100$
$\therefore 55 + 3a = 100$
$\therefore 3a = 45$

$$\boxed{a = 15}$$

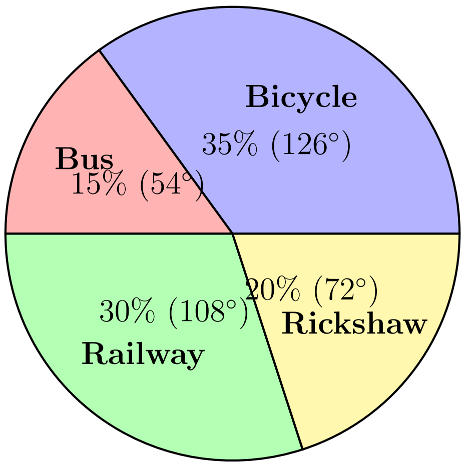
$\therefore$ Bus = 15%, Railway = $2 \times 15 = 30$%

Step 2: Calculate the central angles.

$$\text{Central angle} = \frac{\text{Percentage}}{100} \times 360^\circ$$

Mode of Transport	Percentage	Central Angle
Bicycle	35	$\frac{35}{100} \times 360° = 126°$
Bus	15	$\frac{15}{100} \times 360° = 54°$
Railway	30	$\frac{30}{100} \times 360° = 108°$
Rickshaw	20	$\frac{20}{100} \times 360° = 72°$
Total	**100**	**360°**

Step 3: Draw the Pie diagram.



(ii) There are some students in group A and group B. If one student goes from group A to group B, then the number of students in both groups is same. But one student goes from group B to group A, then the number of students in group A becomes twice of group B. Find the number of students in each group.

Let the number of students in group A be x and in group B be y.

Condition I: If one student goes from group A to group B, the numbers become equal.

$$x - 1 = y + 1 \tag{5}$$

$\therefore x - y = 2 \quad \ldots \text{(I)}$

Condition II: If one student goes from group B to group A, group A becomes twice of group B.

$$x + 1 = 2(y - 1) \tag{6}$$

$\therefore x + 1 = 2y - 2$

$\therefore x - 2y = -3 \quad \ldots \text{(II)}$

Subtracting equation (II) from equation (I):

$(x - y) - (x - 2y) = 2 - (-3)$

$\therefore x - y - x + 2y = 5$

$\therefore y = 5$

Substituting $y = 5$ in equation (I):
$x - 5 = 2$
$\therefore x = 7$

$$\boxed{\text{Number of students in group A} = 7, \quad \text{group B} = 5}$$

Verification:

- If 1 student goes from A to B: A = 6, B = 6 (Equal) ✓
- If 1 student goes from B to A: A = 8, B = 4 ($8 = 2 \times 4$) ✓

— End of Solutions —